

Early Gradient Spectra Predict Useful Low-Rank Adaptation

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Abstract

LoRA and related PEFT methods expose a nominal rank, but the rank that is actually useful depends on the task spectrum seen through the model’s activations. We study this with a reduced-rank adaptation model and controlled synthetic experiments. For a frozen map with activation covariance C and task update Δ_* , the best rank- r adapter is the truncated SVD of $M_* = \Delta_* C^{1/2}$, and its excess risk is $\frac{1}{2} \sum_{i>r} \sigma_i^2(M_*)$. Moreover, the population early gradient, after activation whitening, has exactly this spectrum. Thus early-gradient spectra can predict useful adapter rank before training. We evaluate this prediction in multi-seed matrix simulations using useful-rank targets normalized by the best validation loss observed on the tested rank grid; the simulation oracle is diagnostic and does not set any target. We then implement the estimator in small transformer modules on synthetic modular-arithmetic and associative-recall adaptation tasks. Across six run-level clusters (three seeds per task, yielding 39 descriptive task-budget evaluations per rule), early-gradient allocation rules outperform a fair uniform-fill LoRA rank baseline on most budgets; the strongest spectral allocator, a soft-threshold spectral dimension, improves validation loss by 0.107 on average, wins on 33/39 descriptive task-budget cases, and has negative mean loss delta in 5/6 run clusters. A whitening ablation shows that activation weighting is important for effective rank, while soft-threshold dimensions are robust across whitening choices. The results support a cautious claim: early-gradient spectra identify useful low-rank task directions and provide practical signals for where LoRA rank should be spent. A small fixed-scale GPT-2/Wikitext-2 validation gives a limited real-model check: spectral-effective allocation has a small mean advantage over uniform rank 4 in a restricted target-module setting, while an attention-output projection ablation exposes a module-family boundary condition.

Scope and claim boundary. Most experiments are synthetic and theorem-driven; the final validation uses GPT-2 on Wikitext-2 as a small real pretrained-transformer check. We do not claim that nominal rank predicts performance, that the unconstrained best rank is predictable, that any single spectral rule dominates all baselines, or that the present theory proves nonlinear transformer convergence. The intended contribution is a principled spectral predictor of useful rank and controlled evidence that it yields useful rank-allocation signals in matrix, synthetic-transformer, and one small real-transformer setting.

1 Introduction

Low-rank adaptation methods such as LoRA make rank a visible and tunable resource. A rank- r LoRA adapter learns

$$\Delta W = \frac{\alpha}{r} B A, \quad (1)$$

with $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $A \in \mathbb{R}^{r \times d_{\text{in}}}$. In practice, however, nominal rank is an incomplete description of adaptation capacity. Rank interacts with scale, optimization time, activation geometry, finite-sample noise, and the module on which the adapter is placed. A rank that is algebraically available may not be functionally useful, and a small budget spent on the right module can outperform a larger budget spread uniformly.

We separate *nominal rank* from *useful rank*. Nominal rank is the algebraic rank supplied to the adapter. Useful rank is the smallest rank satisfying a practical criterion: near-best validation loss, a fixed fraction of attainable improvement, or a rank-penalized tradeoff. Our central observation is that useful rank is a spectral quantity in function space. If C is the activation covariance and Δ_* is the task update, the relevant operator is

$$M_* = \Delta_* C^{1/2}. \quad (2)$$

This activation weighting discounts update directions that the model rarely uses.

We prove two exact population statements in a reduced-rank adaptation model. First, the best rank- r adapter is obtained by truncating the singular spectrum of M_* . Second, the early full gradient at the frozen model, after activation whitening, equals $-M_*$. Therefore, the useful-rank spectrum is visible before training an adapter. We test this idea in controlled matrix simulations and then in small synthetic transformer modules. The transformer experiments show that early-gradient spectral statistics are useful not only for predicting rank in an isolated matrix problem, but also for deciding where a limited LoRA rank budget should be spent across attention and MLP modules.

Contributions.

1. We give an exact reduced-rank adaptation theorem: rank- r excess risk is the tail singular energy of $\Delta_* C^{1/2}$.
2. We show that the activation-whitened population early gradient has the same spectrum as this task operator.
3. We define useful-rank targets—near-best, recovery, and penalized rank—that avoid the degeneracy of unconstrained best rank.

4. We validate useful-rank prediction in multi-seed spiked matrix simulations, where gradient effective rank strongly predicts practical rank targets.
5. We implement the estimator in small transformer modules and show that early-gradient allocation rules beat fair uniform-fill LoRA rank allocation across synthetic modular-arithmetic and associative-recall adaptation tasks, with whitening ablations supporting the activation-weighted view.
6. We add a small real GPT-2/Wikitext-2 validation showing a small descriptive mean advantage over uniform rank 4 and a larger advantage over gradient norm in a restricted target-module setting, while an attention-output projection ablation reveals a boundary condition.

2 Related Work

LoRA and PEFT. LoRA freezes pretrained weights and trains low-rank increments, giving strong fine-tuning performance with far fewer trainable parameters [1]. QLoRA backpropagates through quantized frozen models into LoRA adapters, making LoRA-style adaptation central to memory-efficient fine-tuning [2]. Our focus is orthogonal: how much low-rank capacity is useful, and how should it be allocated?

Adaptive rank and scaling. AdaLoRA explicitly re-allocates rank budget across weight matrices using importance scores [3]. Rank-stabilized LoRA changes the standard α/r scaling to mitigate high-rank learning issues [4]. DoRA decomposes weights into magnitude and direction and applies low-rank updates to the direction component [5]. These works motivate rank allocation and rank-scale disentanglement. We contribute a pre-training spectral predictor of rank demand.

Gradient subspaces and spectral perturbation. LoRA-One studies one-step full-gradient subspaces and shows that singular directions of the gradient can guide low-rank adaptation [6]. We ask a complementary question: not only which subspace, but how many directions. Mathematically, we build on Eckart–Young–Mirsky low-rank approximation [8, 9], classical perturbation theory [10], and spiked random-matrix transitions [13]. Pythia is a natural future open-model validation suite because it provides controlled checkpoints across scale and training time [7]; we do not use it in this paper.

3 Theory

Let

$$\begin{aligned} y &= (W_0 + \Delta_\star)x + \xi, \\ \mathbb{E}[x] &= 0, \quad \mathbb{E}[xx^\top] = C \succ 0, \quad \mathbb{E}[\xi \mid x] = 0. \end{aligned} \quad (3)$$

A rank-constrained adapter predicts $(W_0 + \Delta)x$ with $\text{rank}(\Delta) \leq r$, and

$$\mathcal{L}(\Delta) = \frac{1}{2} \mathbb{E} \|y - (W_0 + \Delta)x\|_2^2. \quad (5)$$

Define $M_\star = \Delta_\star C^{1/2}$.

Theorem 3.1 (Population reduced-rank adaptation). *Let $M_\star = U\Sigma V^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$. The best rank- r adapter is*

$$\Delta_r^\star = U_r \Sigma_r V_r^\top C^{-1/2}, \quad (6)$$

and its excess risk is

$$\mathcal{L}(\Delta_r^\star) - \mathcal{L}(\Delta_\star) = \frac{1}{2} \sum_{i>r} \sigma_i^2(M_\star). \quad (7)$$

The smallest rank recovering fraction ρ of the attainable improvement is

$$r_{\text{rec}}(\rho) = \min \left\{ r : \sum_{i \leq r} \sigma_i^2(M_\star) \geq \rho \sum_i \sigma_i^2(M_\star) \right\}. \quad (8)$$

Proposition 3.2 (Early-gradient spectrum). *Let $G_0 = \nabla_\Delta \mathcal{L}(0)$. Then*

$$G_0 = -\Delta_\star C, \quad -G_0 C^{-1/2} = \Delta_\star C^{1/2} = M_\star. \quad (9)$$

Thus the activation-whitened early gradient has the spectrum that determines useful rank.

Predictors. We compare hard detectable rank, stable rank, effective rank, and soft spectral dimensions. The strongest replicated predictor is effective rank:

$$r_{\text{eff}}(A) = \exp \left(- \sum_i p_i(A) \log p_i(A) \right), \quad p_i(A) = \frac{\sigma_i^2(A)}{\|A\|_F^2}. \quad (10)$$

A threshold-aware continuous alternative is

$$d_{\phi, \tau}(A) = \sum_i \phi \left(\frac{\sigma_i^2(A)}{\tau^2} \right). \quad (11)$$

Hard detectable rank corresponds to the discontinuous gate $\phi(t) = \mathbf{1}\{t > 1\}$.

Finite samples. With empirical gradient $\hat{G}_0$, ideal whitening gives $\hat{M} = -\hat{G}_0 C^{-1/2} = M_\star + E$. If $\|E\|_{\text{op}} \leq \varepsilon$, Weyl’s inequality controls singular values. Hard thresholding is stable only with a gap around the edge; effective rank is continuous under relative Frobenius perturbation. In practice we use regularized empirical whitening:

$$\hat{M}_\lambda = -\hat{G}_0 (\hat{C} + \lambda I)^{-1/2}. \quad (12)$$

4 Experimental Setup

The matrix and small-transformer experiments are synthetic and require no external datasets. The final GPT-2/Wikitext-2 validation uses local pretrained-model and dataset files to test whether the diagnostic survives one real pretrained-transformer setting. The goal is to test the rank-prediction mechanism under controlled task spectra, sample noise, rank grids, scale choices, and a small real-model check. Table 1 summarizes the experimental regimes.

Synthetic regression. Tasks use $y = (W_0 + \Delta_\star)x + \xi$. Default dimensions are $d_{in} = d_{out} = 128$ with power-law activation covariance. The main task-rank-8 spike values are

$$(2.8, 2.2, 1.8, 1.45, 1.1, 0.85, 0.65, 0.5). \quad (13)$$

LoRA factors are trained by gradient descent with clipping. Detection thresholds use a permutation null at quantile 0.995.

Useful-rank targets. Let $\mathcal{R}$ be the tested nominal-rank grid. For validation curve $V(r)$, define $V_0 = V(0)$, $V_{\min} = \min_{r \in \mathcal{R}} V(r)$, $\Delta V = V_0 - V_{\min}$, and $Q(r) = (V_0 - V(r))/\Delta V$. When $\Delta V > 0$, we report

$$\begin{aligned} r_{\text{near}}(\gamma) &= \min\{r \in \mathcal{R} : V(r) \leq V_{\min} + \gamma\Delta V\}, \\ r_{\text{rec}}(\rho) &= \min\{r \in \mathcal{R} : Q(r) \geq \rho\}, \\ r_{\text{pen}}(\lambda) &= \operatorname{argmin}_{r \in \mathcal{R}} \{V(r) + \lambda\Delta V r/r_{\max}\}. \end{aligned}$$

Ties are resolved in favor of the smallest nominal rank. Curves with $\Delta V \leq 0$ have undefined useful-rank targets and are excluded from target fits. The simulation oracle loss is retained only as a diagnostic and never enters $V_{\min}$, ΔV , a threshold, or a penalty. The unconstrained observed-best rank is likewise diagnostic; in layerwise experiments it is often the largest rank tested.

5 Results

5.1 Detectability and nominal-rank failures

The BBP-style simulation confirms the expected detectability transition (Figure 1, left). In rank-1 Gaussian noise with edge near 2.0, planted/empirical overlap is near zero for small spikes and rises after spectral separation: mean overlap is about 0.008 at $\theta = 0.1$, 0.288 at $\theta = 1.2$, 0.540 at $\theta = 1.5$, and 0.893 at $\theta = 3.0$.

The rank-sweep panels in Figure 1 show why nominal rank alone is insufficient. A cleaned fixed- α LoRA sweep with $\alpha = 16$, learning rate 0.2, 1200 steps, ranks 1–64, and 8 repetitions shows a U-shaped curve. Mean validation loss is 0.1011 at rank 1, 0.0211 at rank 4, 0.0058 at rank 8, 0.0174 at rank 16, and 0.1303 at rank 64. Robust fits show that final adapter stable/effective/detectable rank explain validation loss far better than nominal rank.

5.2 Rank and scale are confounded

Alpha controls both optimization and drift. In the alpha stress run, validation loss improves up to moderate α and then degrades mildly, while the output-drift proxy increases. Large α did not robustly create separate “intruder” directions in this synthetic setting; the safer interpretation is amplification and drift. A rank-alpha grid further shows that nominal rank alone is weak: on the clean subset with validation loss < 1 , adapter stable rank reaches $R^2 \approx 0.810$ for log validation loss, adapter detectable rank plus alpha reaches $R^2 \approx 0.778$, adapter effective rank reaches $R^2 \approx 0.740$, alpha alone reaches $R^2 \approx 0.552$, and nominal rank alone reaches only $R^2 \approx 0.040$.

5.3 Signed conflict is secondary but informative

The original unsigned merge-overlap experiment was weak: spectral interference explained only about $R^2 = 0.051$ for merge degradation and planted overlap about $R^2 = 0.046$. A corrected signed-conflict experiment gave a clearer story: the signed task inner product explains merge degradation with $R^2 = 0.9995$, while a conflict score gives $R^2 = 0.747$ and unsigned overlap alone gives $R^2 = 0.085$. This is a secondary result; it supports using signed spectral geometry rather than unsigned overlap for task-vector interference.

5.4 Stage4: early-gradient effective rank predicts useful rank

The main matrix-simulation evidence is the Stage4 replication: five seeds per condition and 48 synthetic layers per seed. For each layer, we compute activation-whitened early-gradient spectral statistics, run rank sweeps over $\{1, 2, 4, 8, 16, 32\}$, and evaluate useful-rank targets. Before the corrected rerun, we designate the hard-knee $r_{\text{near}}(0.1)$ versus activation-whitened gradient effective rank comparison as the primary confirmatory pair; its sample-limited counterpart is a prespecified robustness check. Other displayed targets and all alternative predictors are secondary or exploratory. Figure 2 and Table 2 report the replicated activation-whitened effective-rank predictor.

The hard-knee regime uses $n_{\text{train}} = 1024$, label noise 0.05, and weak-factor range 0.01–0.08. The sample-limited regime uses $n_{\text{train}} = 384$, label noise 0.12, and weak-factor range 0.05–0.25. The unconstrained best rank is degenerate: rank 32 is best for almost every layer. Useful-rank targets are nondegenerate and predictable.

Table 1: Experimental settings. Stage4 is the primary evidence for early-gradient useful-rank prediction.

Experiment	Regime	Sweep	Purpose
BBP	$m = n = 128$, rank-1 spike	spike strength θ	verify spectral detectability transition
LoRA rank	$d_{in} = d_{out} = 128$, task rank 8	ranks 1–64	test nominal rank and final adapter spectra
Alpha / rank-alpha Merge/conflict	noisy or rank-8 tasks paired task vectors	α and rank grids overlap/sign sweeps	separate rank, scale, and instability test signed versus unsigned interference
Layerwise hard-knee	48 synthetic layers	ranks 1–32	test early-gradient useful-rank prediction
Layerwise sample-limited	fewer samples and more noise	ranks 1–32	test finite-sample degradation
Stage4 replication	5 seeds $\times$ 2 regimes	48 layers per seed	multi-seed validation

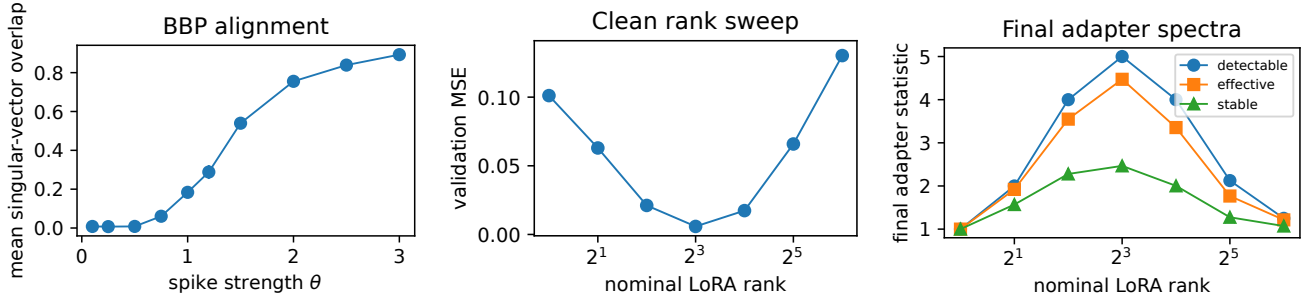


Figure 1: Left: BBP detectability transition. Middle: a clean fixed- α LoRA rank sweep is U-shaped, so nominal rank is not the explanatory variable. Right: final adapter spectral statistics peak near the useful-rank range.

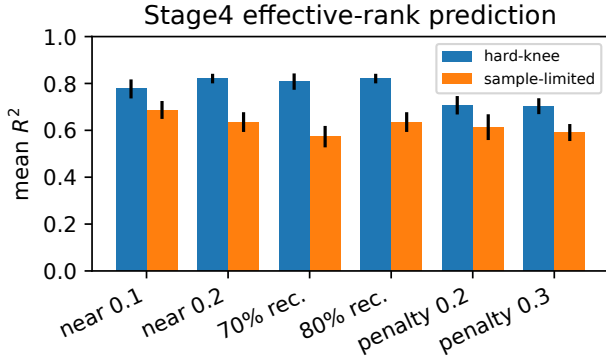


Figure 2: Stage4 replication. Bars show mean R^2 over 5 seeds for activation-whitened gradient effective rank predicting $\log_2$ useful-rank targets. Hard-knee spectra are strongly predicted; sample-limited spectra are weaker but still informative.

Table 2: Stage4 results for activation-whitened gradient effective rank. Mean over five seeds; SEM in parentheses.

Condition	Target	Mean R^2	Mean Spearman
Hard-knee	near-best 0.1 gap	0.777 (0.041)	0.852 (0.017)
Hard-knee	near-best 0.2 gap	0.821 (0.020)	0.868 (0.013)
Hard-knee	70% recovery	0.808 (0.035)	0.872 (0.019)
Hard-knee	80% recovery	0.821 (0.020)	0.868 (0.013)
Hard-knee	penalty 0.2	0.707 (0.039)	0.804 (0.030)
Hard-knee	penalty 0.3	0.703 (0.034)	0.811 (0.023)
Sample-limited	near-best 0.1 gap	0.687 (0.038)	0.786 (0.026)
Sample-limited	near-best 0.2 gap	0.635 (0.042)	0.766 (0.030)
Sample-limited	70% recovery	0.573 (0.046)	0.731 (0.039)
Sample-limited	80% recovery	0.635 (0.042)	0.766 (0.030)
Sample-limited	penalty 0.2	0.614 (0.055)	0.740 (0.040)
Sample-limited	penalty 0.3	0.591 (0.036)	0.720 (0.019)

6 Synthetic-Transformer Module Experiments

We next test whether the spectral objects above are useful inside actual transformer modules. The data remain fully synthetic, so the setting is controlled and reproducible. Each experiment trains a small two-layer decoder-only transformer on a base sequence task, freezes it, and adapts to a shifted task using LoRA. The model uses $d_{\text{model}} = 64$, four attention heads, MLP width 128, and no dropout. Candidate adapter sites are attention projections and MLP output projections.

Calibration pipeline. For each candidate site j , forward and backward hooks collect module inputs $x_{j,b,a}$ and output-gradient vectors $\tilde{g}_{j,b,a}$ from PyTorch backpropagation of the mean token loss on calibration batch b . The implementation estimates

$$\hat{C}_j = \frac{1}{N_j} \sum_{b,a} x_{j,b,a} x_{j,b,a}^\top, \quad \hat{G}_j = \frac{1}{B_j} \sum_b \sum_a \tilde{g}_{j,b,a} x_{j,b,a}^\top, \quad (14)$$

where $\tilde{g}$ already includes the batch mean-loss normalization. This is the empirical full-gradient estimator used by the released code. We then form the regularized whitened early-gradient matrix

$$\widehat{M}_{j,\lambda} = -\hat{G}_j(\hat{C}_j + \lambda_j I)^{-1/2}. \quad (15)$$

The runs below use diagonal whitening with $\lambda_j = 0.01 \text{tr}(\hat{C}_j)/d_j$, eight calibration batches of size 128, and a permutation null at quantile 0.995 for thresholded quantities. From the singular values of $\widehat{M}_{j,\lambda}$ we compute effective rank, hard detectable rank, soft spectral dimension, gradient norm, and marginal singular-energy scores.

Tasks and LoRA sweeps. We evaluate two adaptation tasks. In modular arithmetic, the base task uses modulus 31 with offset 0 and the adaptation task shifts to offset 5; base training uses 600 steps. Candidate sites are q , v , o attention projections and MLP $fc2$ in each of two layers. In associative recall, the model uses 16 keys and four key-value pairs; the adaptation task shifts the key-value map by 3; base training uses 1000 steps. Candidate sites additionally include k projections. The runner records base-task and shifted-task accuracy before LoRA in each run’s `base/base_metrics.csv`; the reported summaries use runs with near-perfect base-task accuracy and near-zero shifted-task accuracy before LoRA, yielding a clean distribution shift. LoRA sweeps use ranks $\{0, 1, 2, 4, 8, 16\}$, $\alpha = 16$, learning rate 0.01, clipping at 1.0, and 300–350 adaptation steps. We run three seeds per task.

Budgeted allocation. We compare a fair *uniform-fill* baseline against early-gradient allocation rules. Unlike naive uniform rank, uniform-fill spends the available budget by incrementing ranks as evenly as possible, so it is not

artificially rank-zero at small budgets. Spectral and gradient rules allocate ranks under the same parameter budget using effective rank, soft spectral dimension, gradient norm, or marginal singular-energy gains.

Results. Figure 3 and Table 3 show the descriptive task-budget results. Because budgets within a seed are correlated, Table 4 also reports run-clustered means. The best cross-task spectral allocator is the soft-threshold spectral dimension: it improves mean validation loss by 0.1074 relative to uniform-fill, wins on 33/39 descriptive task-budget evaluations, and has negative mean loss delta in 5/6 run clusters. At low budgets $\{512, 1024, 1536, 2048\}$, soft dimension wins on 19/24 evaluations and gives mean loss improvement 0.1425. Gradient norm is a strong non-spectral baseline, winning 32/39 descriptive evaluations with mean accuracy improvement 0.0317. Effective rank is still useful, especially as a sitewise rank-demand statistic, but is not the best budget allocator in these transformer tasks. Marginal-gain edge is the least reliable rule in aggregate.

The transformer evidence shifts the empirical emphasis. In matrix simulations, early-gradient effective rank predicts a smallest useful rank. In transformer modules, the stronger claim is that early-gradient spectral dimensions identify where a limited rank budget should be spent. These are consistent: a high-spectrum module may be so useful that even low rank recovers most of its own attainable gain, while still deserving budget in a multi-site allocation.

Whitening ablation. The theory predicts that the relevant spectrum is activation weighted, not simply the raw weight-gradient spectrum. We therefore run a whitening ablation on one modular seed and one associative-recall seed, recomputing module scores with no whitening, diagonal whitening, and full regularized whitening. This ablation reuses trained base models and compares allocation rules under the same budgets as above. Table 5 and Figure 4 summarize the combined result over 13 task-budget cases. The best rule is full-whitened soft dimension, with mean validation-loss delta -0.2203 versus uniform-fill and wins on 10/13 cases. It also improves over the gradient-norm baseline, whose mean delta is -0.1559 . Diagonal or full whitening substantially improves effective-rank allocation: raw effective rank has mean delta -0.0105 , while full effective rank reaches -0.1766 and diagonal effective rank wins on all 13/13 cases. Soft dimension is more robust: it works well with no, diagonal, and full whitening. The result is not uniform across tasks; in associative recall, unwhitened soft dimension is strongest, while in modular arithmetic full or diagonal whitening is best. Thus whitening matters, especially for effective rank, but the best practical rule remains task dependent.

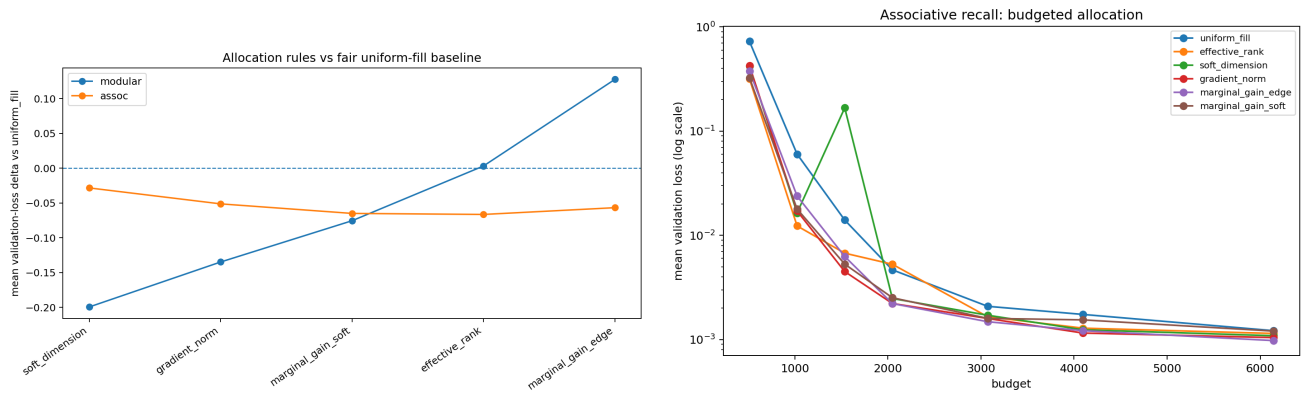


Figure 3: Synthetic-transformer module experiments. Left: mean validation-loss delta relative to the fair uniform-fill baseline; negative is better. Across modular arithmetic and associative recall, early-gradient allocation rules usually improve over uniform-fill. Right: associative recall budget curves on a log-loss scale; early-gradient rules spend small budgets on useful modules earlier than uniform-fill.

Table 3: Synthetic-transformer allocation results across modular arithmetic and associative recall. Deltas are relative to fair uniform-fill; negative loss delta is better. The 39 task-budget evaluations are descriptive because budgets are nested within six run-level clusters: three modular seeds over six budgets and three associative-recall seeds over seven budgets. Low-budget numbers restrict to budgets 512, 1024, 1536, 2048 and 24 evaluations.

Rule	Mean loss Δ	Wins / 39	Low-budget loss Δ	Low-budget wins / 24	Mean acc. Δ
soft dimension	-0.1074	33	-0.1425	19	+0.0300
gradient norm	-0.0899	32	-0.1060	19	+0.0317
marginal gain soft	-0.0699	30	-0.0727	19	+0.0240
effective rank	-0.0343	32	-0.0107	18	+0.0122
marginal gain edge	+0.0284	27	+0.0386	15	-0.0042

Table 4: Run-clustered synthetic-transformer summary. Each run cluster first averages loss deltas over that run’s budgets; the table then summarizes the six run-level means. This is the appropriate sanity check for the nested budget curves in Table 3.

Rule	Mean Δ	SEM	Runs better
soft dimension	-0.1140	0.0596	5/6
gradient norm	-0.0931	0.0510	4/6
marginal gain soft	-0.0703	0.0399	5/6
effective rank	-0.0317	0.0685	5/6
marginal gain edge	+0.0355	0.0641	3/6

Table 5: Whitening ablation on one modular seed and one associative-recall seed. Deltas are relative to fair uniform-fill; negative is better. Low-budget numbers restrict to budgets 512, 1024, 1536, 2048 and eight cases.

Rule	Whitening	Mean loss Δ	Wins / 13	Low-budget loss Δ	Low-budget wins / 8	Mean acc. Δ
soft dimension	full	-0.2203	10	-0.2995	5	+0.0400
soft dimension	none	-0.2105	12	-0.2720	7	+0.0445
marginal gain soft	none	-0.2080	10	-0.2615	5	+0.0466
soft dimension	diag	-0.1856	10	-0.2394	6	+0.0378
effective rank	full	-0.1766	12	-0.2520	7	+0.0427
gradient norm	—	-0.1559	10	-0.2201	5	+0.0247
marginal gain soft	diag	-0.1555	9	-0.1961	5	+0.0365
effective rank	diag	-0.1282	13	-0.1466	8	+0.0358
marginal gain soft	full	-0.0658	7	-0.0695	5	+0.0083
effective rank	none	-0.0105	10	+0.0589	5	-0.0047

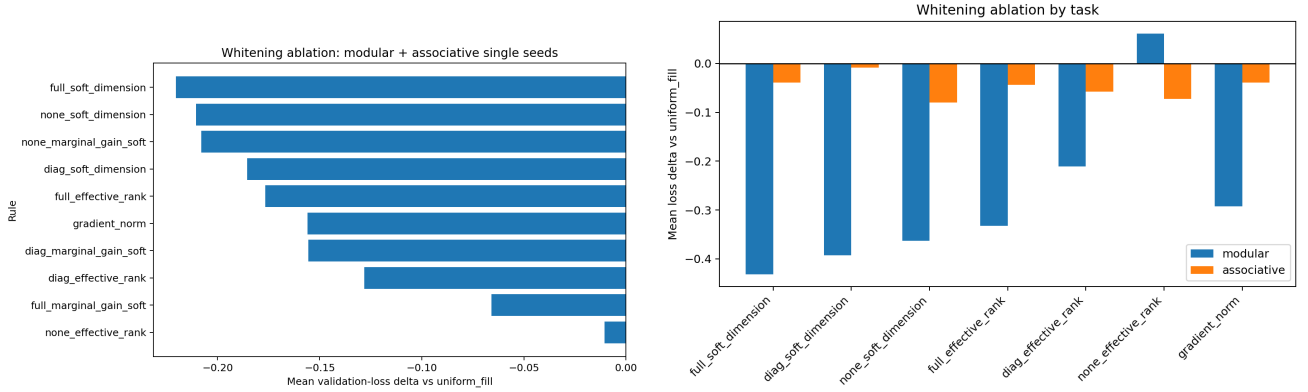


Figure 4: Whitening ablation. Left: combined mean validation-loss delta relative to uniform-fill; negative is better. Right: task-specific mean deltas for selected rules. Whitening improves effective rank strongly on the modular task, while soft dimension remains robust across whitening choices.

7 Real GPT-2/Wikitext Validation

The synthetic experiments isolate the mechanism, but the practical motivation is LoRA allocation in pretrained transformers. We therefore add a small real-model validation using GPT-2 on Wikitext-2 [11, 12]. The experiment freezes a local pretrained GPT-2 checkpoint, calibrates early gradients on local Wikitext-2 blocks, and trains adapters for 200 steps under approximately matched trainable-parameter budget caps. We compare uniform rank-4, gradient-norm allocation, and activation-whitened spectral effective-rank allocation. Candidate modules are GPT-2 Conv1D sites; all runs use the same local model, local dataset files, block size 128, batch size 4, eight calibration batches, and 32 evaluation batches. To isolate allocation from rank-dependent scale, this validation uses a fixed update scale (the released code’s `lora_alpha_scale=2.0`) rather than standard α/r scaling; scaling-rule ablations are future work.

Table 6 shows a limited positive real-model signal in the c_{attn}, c_{fc} setting. Spectral-effective allocation has the lowest mean final validation loss, wins four of five seeds, and uses the same mean trainable-parameter count as uniform rank-4. Its mean loss is lower than uniform by 0.0015 (paired-difference SEM 0.0013) and lower than gradient norm by 0.0475 (paired-difference SEM 0.0010). Given only five seeds and the small separation from uniform, we treat the uniform comparison as descriptive rather than decisive. The allocation is structurally different from gradient norm: gradient norm spends rank on early high-norm attention modules, whereas spectral-effective allocation shifts capacity toward MLP input projections with larger activation-whitened effective rank.

The attention-output projection ablation is deliberately included as a boundary condition. When `attn.c.proj` modules are added, spectral-effective allocation still improves over gradient norm by 0.0317 mean validation loss, but uniform rank-4 wins all three seeds and is better by 0.0051 on average. The calibration reveals why: `attn.c.proj` modules are cheap per rank and have high whitened effective rank, so a naive cost-aware scalar spectral allocator over-allocates to that module family. This argues for group-constrained or module-family-normalized spectral allocation, not for dropping the spectral signal.

8 Rank Allocation Algorithm

The transformer experiments implement the following plug-in rule. For every candidate site j , compute the regularized whitened early gradient $\widehat{M}_{j,\lambda}$. Let S_j be a scalar spectral score such as $r_{\text{eff}}(\widehat{M}_{j,\lambda})$ or $d_{\phi, \widehat{\tau}_j}(\widehat{M}_{j,\lambda})$. A simple allocator chooses

$$r_j(\kappa) = \Pi_{\mathcal{R}_j}(\kappa S_j), \quad \sum_j c_j r_j(\kappa) \leq B, \quad (16)$$

where $c_j = d_{j,\text{in}} + d_{j,\text{out}}$ and κ is calibrated to the budget. A marginal-gain allocator instead estimates gains

$$\widehat{b}_{j,i} = \frac{1}{2} \widehat{\sigma}_{j,i}^2 \quad \text{or} \quad \frac{1}{2} (\widehat{\sigma}_{j,i}^2 - \widehat{\tau}_j^2)_+, \quad (17)$$

and includes directions with $\widehat{b}_{j,i}/c_j > \lambda$. The population version of this rule follows directly from Theorem 3.1: each singular direction reduces risk by $\frac{1}{2} \sigma_{j,i}^2$ at cost c_j .

The empirical takeaway is not that a single rule is universally optimal. Rather, early-gradient rules form a useful allocation family. Soft spectral dimension is the strongest spectral budget allocator in our two transformer tasks, effective rank is the most direct theorem-derived soft-rank statistic, and gradient norm is a competitive non-spectral baseline that must be retained.

9 Future Validation Targets

The synthetic-transformer bridge is positive but limited. The whitening ablation supports the activation-weighted viewpoint, while also showing task dependence: whitening helps effective rank strongly in modular arithmetic, whereas associative recall is already well served by unwhitened or softly thresholded spectra. The next task-level validation should add copy/reverse-copy, in-context regression or classification, and finite-state sequence shifts. We should also test whether the soft-dimension advantage persists when candidate sites include all attention and MLP projections rather than a hand-picked subset, and replicate the whitening ablation across all seeds. A later open-model experiment should use a small checkpoint suite; the present real-model result is treated only as a limited fixed-scale check, not as broad empirical validation.

10 Discussion

The main result is not that nominal rank works; it is that useful adaptation is governed by spectra of activation-weighted task directions. Early gradients estimate those spectra in the reduced-rank model. Effective rank is strongest for matrix useful-rank prediction, while soft-threshold spectral dimension is strongest for budgeted transformer-module allocation in the current experiments.

Limitations. The exact theorem is linear reduced-rank adaptation, not full nonlinear transformer fine-tuning. Most experiments are synthetic, and the real validation is deliberately small: GPT-2 on Wikitext-2 with a restricted set of LoRA target families and fixed update scale. Gradient norm is a strong baseline, uniform rank can win when module families with very different per-rank costs are mixed, and the best spectral statistic is task-dependent. The intruder-direction hypothesis and unsigned merge-overlap hypothesis are not central because the simulations did not support them robustly.

Table 6: Real GPT-2/Wikitext-2 LoRA validation. Final validation loss is mean over seeds with SEM in parentheses. The c_{attn}, c_{fc} setting uses attention input projections and MLP input projections. The $+a_{proj}$ ablation additionally includes attention output projections.

Setting	Strategy	Final val. loss	PPL	Trainable params	Wins
c_{attn}, c_{fc}	spectral effective	3.4573 (0.0016)	31.73	331776	4/5
c_{attn}, c_{fc}	uniform rank 4	3.4587 (0.0017)	31.78	331776	1/5
c_{attn}, c_{fc}	gradient norm	3.5047 (0.0014)	33.27	330086	0/5
$+a_{proj}$	spectral effective	3.4530 (0.0026)	31.60	404992	0/3
$+a_{proj}$	uniform rank 4	3.4479 (0.0006)	31.44	405504	3/3
$+a_{proj}$	gradient norm	3.4847 (0.0014)	32.61	405248	0/3

Claim boundary. The safest claim is: in reduced-rank adaptation and spiked synthetic LoRA simulations, early-gradient spectral dimensions, especially effective rank, predict the smallest useful adapter rank. In small synthetic transformer modules, early-gradient spectral allocation rules beat fair uniform-fill rank allocation under equal budgets, with soft-threshold spectral dimension the strongest overall spectral allocator in our current tasks. In a small GPT-2/Wikitext-2 validation, spectral-effective allocation transfers in a restricted c_{attn}, c_{fc} setting but not when attention-output projections are added without module-family constraints. We do not claim that one spectral rule uniformly dominates gradient norm or uniform rank across all target-module choices.

11 Conclusion

Useful low-rank adaptation is a spectral problem. In the population reduced-rank model, useful rank is exactly the spectral energy rank of $\Delta_* C^{1/2}$, and the activation-whitened early gradient reveals this spectrum before adapter training. Multi-seed matrix simulations support useful-rank prediction. Synthetic transformer experiments show that early-gradient spectral dimensions can allocate LoRA rank better than fair uniform-fill baselines, and a small GPT-2/Wikitext-2 validation provides a limited pretrained-transformer check. The attention-projection ablation also sets a clear boundary: practical allocators should respect module-family structure rather than spend rank purely by scalar spectral score per parameter.

Reproducibility Statement

All experiments write CSV metrics, configuration files, and figures. The matrix and small-transformer experiments use synthetic data; the real validation uses a local GPT-2 checkpoint and local Wikitext-2 parquet-to-text files. The clean release has two source directories, `Code/` and `Paper/`, plus the compiled top-level manuscript `paper.pdf`. Inside `Code/`, `transformer/` contains the synthetic-transformer module experiments, `rmt_lora_sim/` contains the BBP, LoRA rank/scale, signed-conflict, and Stage4 matrix simulations, and

`real_lora_validation/` contains the GPT-2/Wikitext-2 validation. Released matrix CSV outputs are stored under semantic names in `rmt_lora_sim/results/released/`; new runs write to package-local `runs/` directories. Paper-facing Stage4 rows, real-LoRA rows, figures, and generated table rows are regenerated from released CSVs by `Paper/import_stage4_aggregate.py`, `Paper/import_real_lora_results.py`, and `Paper/make_paper_artifacts.py`. The released real-model tables use normalized squared singular values, $p_i = \sigma_i^2 / \sum_j \sigma_j^2$, for entropy effective rank; the superseded raw-singular-value tables are retained in an explicitly labeled archive, and the corrected per-seed evidence includes input manifests, environment records, and SHA-256 checksums.

A Proofs and Additional Theory

Proof of Theorem 3.1. Using $y = (W_0 + \Delta_*)x + \xi$ and $\mathbb{E}[\xi | x] = 0$,

$$\mathcal{L}(\Delta) - \mathcal{L}(\Delta_*) = \frac{1}{2} \|(\Delta_* - \Delta)C^{1/2}\|_F^2. \quad (18)$$

Let $M = \Delta C^{1/2}$. Because $C^{1/2}$ is invertible, $\text{rank}(M) = \text{rank}(\Delta)$. Hence the rank- r problem is

$$\min_{\text{rank}(M) \leq r} \frac{1}{2} \|M_* - M\|_F^2, \quad (19)$$

whose solution is the truncated SVD.

Finite-sample perturbation. If $\widehat{M} = M_* + E$ and $\|E\|_{\text{op}} \leq \varepsilon$, then $|\sigma_i(\widehat{M}) - \sigma_i(M_*)| \leq \varepsilon$. Hence the number of empirical singular values above a threshold τ is trapped between the number of population singular values above $\tau + \varepsilon$ and the number above $\tau - \varepsilon$. This is the detectable-rank sandwich used to explain why hard thresholds require a spectral gap around the edge.

LoRA factorization bridge. In whitened coordinates write $M = sBA$, $s = \alpha/r$, and $F(M) = \frac{1}{2} \|M - M_*\|_F^2$. The factorized global optimum equals the best rank- r approximation because BA represents any rank- r matrix. With zero- B initialization, one gradient step gives

$$M_1 = \eta s^2 M_* A_0^\top A_0. \quad (20)$$

Thus the first LoRA step is a randomized spectral sketch of the task operator. In aligned singular coordinates, the idealized gradient-flow mode amplitude satisfies

$$\dot{m}_i(t) = 2s m_i(t)(\sigma_i - m_i(t)), \quad (21)$$

so larger singular modes are learned faster.

B Reproducibility Details

The project generated archives for BBP, LoRA rank, alpha, merge, stage1b, stage2, stage3, stage4, and synthetic-transformer experiments. Each run writes CSV metrics and figure files. The transformer runs used a two-layer decoder-only transformer with $d_{\text{model}} = 64$, four heads, MLP width 128, diagonal whitening with $\lambda = 0.01 \text{ tr}(\hat{C})/d$, ranks $\{0, 1, 2, 4, 8, 16\}$, and LoRA $\alpha = 16$. Stage4 fits use

$$\log_2 r_j^{\text{target}} = a + b \log_2(1 + s_j) + \epsilon_j, \quad (22)$$

where s_j is a spectral score. Main targets are $r_{\text{near}}(0.1)$, $r_{\text{near}}(0.2)$, $r_{\text{rec}}(0.7)$, $r_{\text{rec}}(0.8)$, $r_{\text{pen}}(0.2)$, and $r_{\text{pen}}(0.3)$.

Release commands. From `Code/`, `make test` runs the lightweight code tests, `make smoke-transformer` runs the synthetic-transformer smoke path, and `make smoke-rmt` runs the general RMT/LoRA smoke path. Stage4 has a dedicated end-to-end check, `cd rmt_lora_sim && bash scripts/run_stage4_smoke.sh`. The paper-grade Stage4 command is `bash scripts/run_stage4.sh`; it writes all per-rank source rows, an aggregate manifest, and nested SHA-256 manifests to a self-contained release under `runs/stage4_releases/`. After validating that five-seed release, `make stage4-to-paper` imports it. The superseded bundled Stage4 aggregate is intentionally rejected. `make real-lora-to-paper` imports the released GPT-2/Wikitext summaries, and `make paper` regenerates tables and figures, compiles the manuscript in `Paper/`, and copies the PDF to top-level `paper.pdf`.

Table 7 is the task-specific complement to the pooled transformer table. It shows that pooled raw deltas should be read descriptively: modular arithmetic and associative recall have different loss scales, and associative recall contains near-zero uniform-fill losses at some budgets that can inflate ratios even when raw deltas are small.

Table 7: Task-specific transformer allocation summary. Deltas are relative to uniform-fill. This table separates task scales instead of averaging only raw losses across modular arithmetic and associative recall.

Task	Rule	Mean loss Δ	Wins	Mean acc. Δ	Mean loss ratio
Modular	soft dimension	-0.1996	6/6	+0.0465	0.874
Modular	gradient norm	-0.1349	5/6	+0.0407	0.908
Modular	marginal gain soft	-0.0756	4/6	+0.0241	0.945
Modular	effective rank	+0.0032	4/6	+0.0020	0.955
Modular	marginal gain edge	+0.1277	2/6	-0.0370	1.057
Associative recall	soft dimension	-0.0283	6/7	+0.0158	3.167
Associative recall	gradient norm	-0.0513	7/7	+0.0240	0.683
Associative recall	marginal gain soft	-0.0650	7/7	+0.0240	0.728
Associative recall	effective rank	-0.0665	6/7	+0.0210	0.791
Associative recall	marginal gain edge	-0.0568	7/7	+0.0240	0.702

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